Class 24: International Investing I

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Today's Question

Why don’t U.S. investors hold the global market portfolio — and what does it cost them?

Positioning

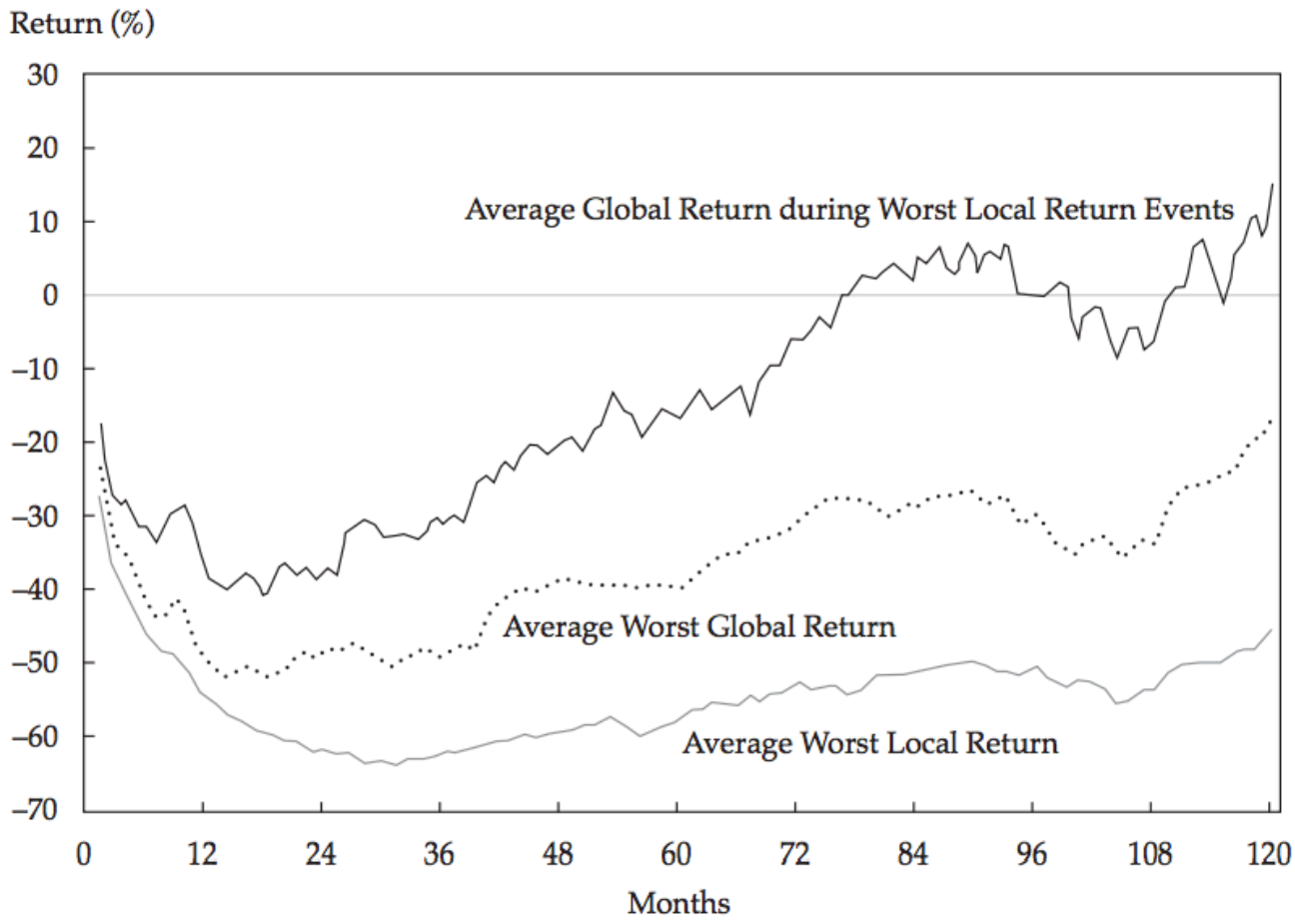
Where We've Been	Where We Are	Where We're Going
Fixed income	Currency risk and international diversification	Emerging markets; global portfolios

Learning Objectives

By the end of today's class, you should be able to:

1. Describe the case for international diversification
2. Explain currency risk and how FX movements affect returns
3. Use covered interest rate parity to price FX forwards and hedge currency exposure
4. Explain the carry trade and uncovered interest rate parity (UIP)
5. Articulate why the carry trade has historically been profitable despite UIP

Long-term Performance of International Diversification



We know that adding assets to a portfolio can reduce variance.

- The CAPM market portfolio included *all* available assets

Only question: how much to weight the assets?

- In equilibrium, market weights are informative
- Apply this logic to international markets

Country	Market Cap (US$ Billions)	% of World	GDP (US$ Billions)	Mkt Cap as % of GDP
World	**136,287**	**100%**	**111,000**	**108%**
United States	69,000	50.6	29,180	219
China	14,750	10.8	18,600	79
Japan	8,080	5.9	4,070	172
Hong Kong	6,490	4.8	407	1,484
India	5,130	3.8	3,890	132
United Kingdom	4,170	3.1	3,730	99
Canada	3,850	2.8	2,240	158
South Korea	3,760	2.8	1,870	201
France	3,690	2.7	3,170	103
Taiwan	3,320	2.4	790	420
Germany	3,150	2.3	4,710	60
Saudi Arabia	2,550	1.9	1,070	238
Switzerland	2,200	1.6	1,130	195
Netherlands	2,100	1.5	1,220	172
Australia	2,090	1.5	2,040	102
Spain	1,960	1.4	1,830	107
Sweden	1,410	1.0	620	227
South Africa	1,230	0.9	420	292
UAE	1,140	0.8	550	207
Brazil	970	0.7	2,330	42

Table 1:

Sources: World Bank Open Data (CM.MKT.LCAP.CD); World Federation of Exchanges, monthly statistics through Dec 2025; IMF World Economic Outlook (GDP).
Notes: Figures rounded to the nearest $10 B for major markets. Hong Kong's market-cap-to-GDP ratio is inflated by its small domestic GDP base relative to its role as a listing hub for mainland Chinese firms.

Home Bias

	Portfolio Weight			Adj. Market Value
	U.S.	Japan	U.K.	
U.S.	.938	.0131	.059	$2941.3
Japan	.031	.9811	.048	1632.9
U.K.	.011	.0019	.820	849.8
France	.005	.0013	.032	265.4
Germany	.005	.0013	.035	235.8
Canada	.010	.0012	.006	233.5

Note: Estimates correspond to portfolio holdings in December, 1989. They are based on the authors' tabulations using data from the *U.S. Treasury Bulletin* and Michael Howell and Angela Cozzini (1990). Adjusted

Home Bias (continued)

True in more recent data too:

- U.S. investors held **78%** of their equity portfolio in U.S. stocks
- Canadian investors held **76%** in Canadian stocks
- German investors held **51%** in German stocks
- U.K. investors held **61%** in U.K. stocks

Improvement over earlier decades, but still lots of room for growth.

Is the International "Market Portfolio" Sensible?

There are reasons *not* to hold the global “market”:

- In many jurisdictions, taxes and other regulations favor local investors over foreign investors
  - This will distort equilibrium market weights
- CAPM assumes all investors face *identical* expected returns and covariances
  - What about exchange rate volatility? Inflation?
  - If we consume in local currency, investors in different countries may face different optimization problems

Impact of FX on Returns

Consider the impact of exchange rate movement on dollar-denominated stock returns:

Domestic Currency Returns (1999-2010)	Australia	France	Germany	Hong Kong	Japan	UK	US
Avg Return	5.24%	0.71%	3.75%	10.31%	-1.11%	-0.05%	-0.31%
Standard deviation	13.33%	19.23%	24.01%	24.59%	20.17%	14.81%	15.90%

USD Returns (1999-2010)	Australia	France	Germany	Hong Kong	Japan	UK	US
Avg Return	9.99%	3.03%	6.19%	10.30%	0.87%	0.09%	-0.31%
Standard deviation	19.49%	21.12%	25.78%	24.66%	20.35%	16.81%	15.90%

Foreign currency appreciation (USD depreciation) has bolstered foreign returns over this decade.

Hedging Exchange Rate Risk

Hedging exchange rate risk is straightforward given a currency futures market when returns are riskless.

- More difficult when returns are **risky**

To do so, we need to understand the basics of forward and futures contracts.

Forwards and Futures

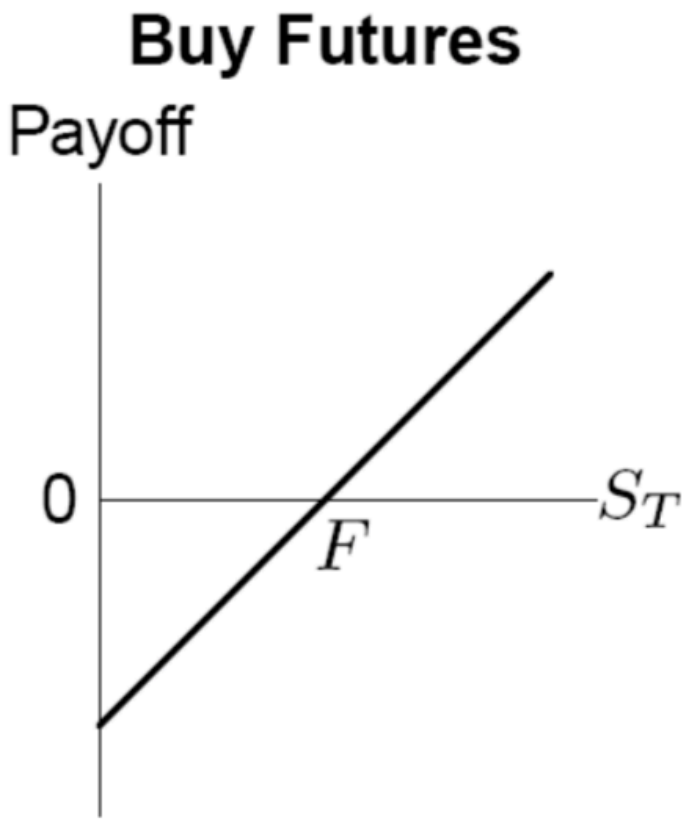
A **forward contract** is a commitment to purchase or sell at a future date a given amount of an asset at a price agreed upon today.

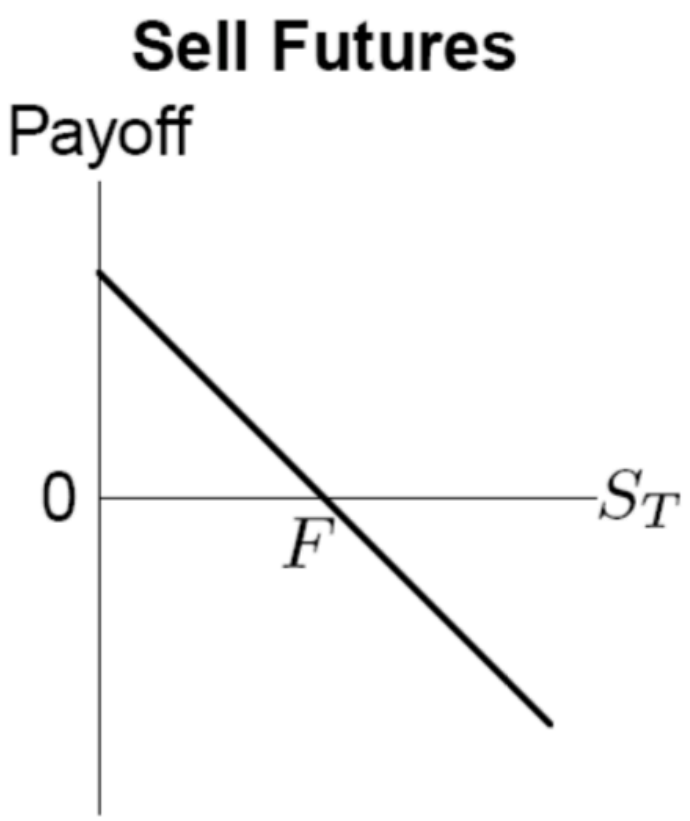
- Currencies, commodities, equities/indices, rates

- F_0 is the **forward price** and represents the fixed price (agreed at $t = 0$) for a future exchange
- S_0 represents the **spot price** prevailing today; S_T represents the spot price at future date T

Note: both parties are bound to the contract — there is no option value.

Forwards and Futures Payoff





Forwards and Futures Payoff

To **buy** a future/forward contract is to be **long** on the asset:

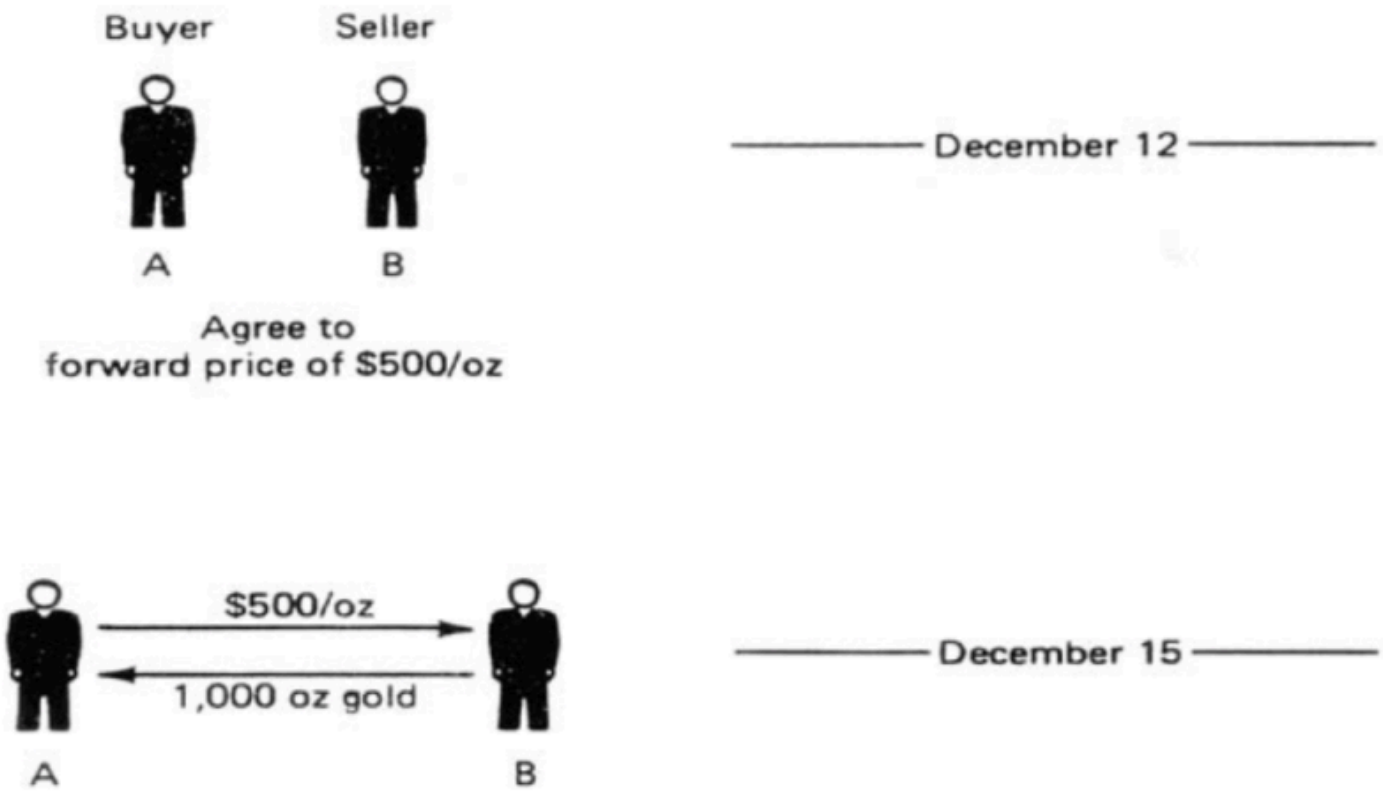
- Obligation to buy
- Payoff is $S_T - F_0$

To **sell** a future/forward contract is to be **short** on the asset:

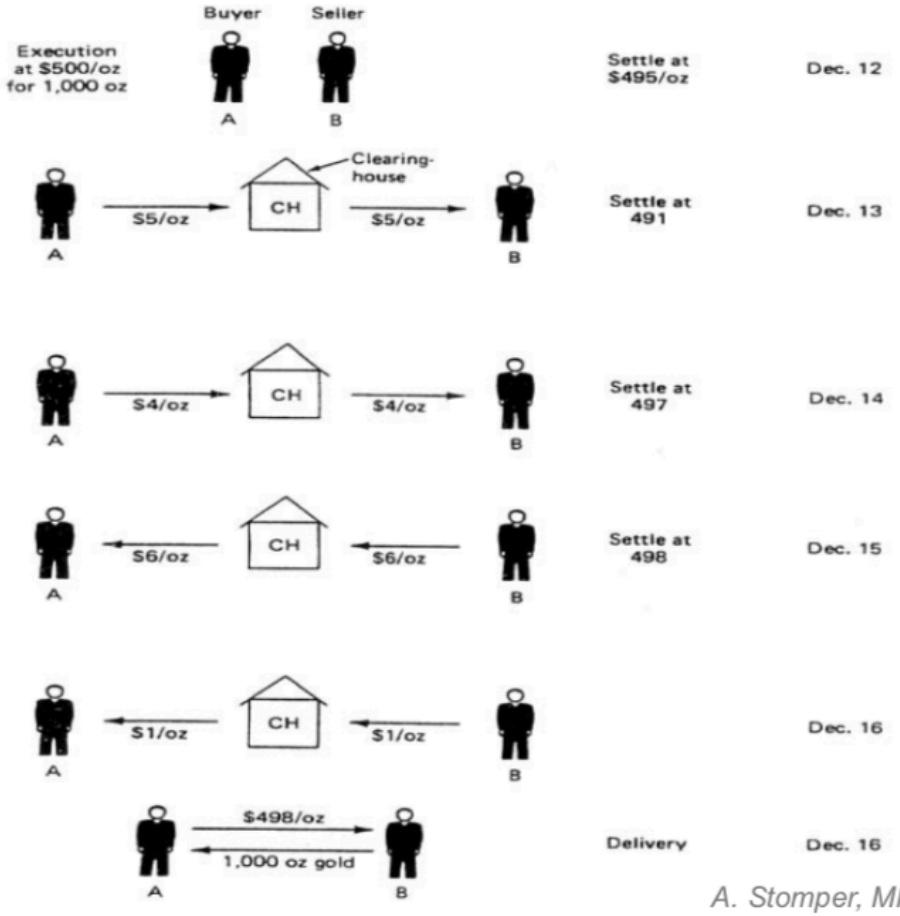
- Obligation to sell
- Payoff is $F_0 - S_T$

Key differences between forwards and futures...

Forward Transaction



Future Transaction



A. Stomper, MIT, 2012

Pricing FX Forwards/Futures

Consider the following strategy:

1. Buy 1 Australian Dollar at S_0 (USD/AUS), financed at $(1 + r_{\text{US}})$
2. Invest in risk-free Australian bond at $(1 + r_{\text{AUS}})^T$
3. Sell $(1 + r_{\text{AUS}})^T$ futures contracts at futures price F_0 (USD/AUS)

Action	Initial cash flow	End of period cash flow
Borrow S_0	S_0	$-S_0(1+r_{US})^T$
Buy 1 AUD, invest in bonds	$-S_0$	$S_T(1+r_{AUS})^T$
Enter short futures position		$(1+r_{AUS})^T(F_0-S_T)$
TOTAL	0	$F_0(1+r_{AUS})^T-S_0(1+r_{US})^T$

FX Forwards: Covered Interest Rate Parity

This implies:

$$F_{0,T} = S_0 \left(\frac{1 + r_{\text{local}}}{1 + r_{\text{foreign}}} \right)^T$$

where the spot price is quoted in **local currency per unit of foreign currency**.

This relationship is called **covered interest rate parity**:

- A hedged investment in risk-free foreign bonds only earns the local risk-free return
- Should hold exactly by no-arbitrage

Question: What happens if you *don't* hedge?

The Carry Trade

The **carry trade** borrows in low-interest-rate currencies and lends in high-interest-rate currencies.

Profitable if either:

1. Exchange rates remain unchanged
2. The high-interest-rate currency appreciates

Alternatively, use the **futures market** instead of borrowing and lending:

- Buying futures is equivalent to borrowing S_0 locally and lending in the foreign market
- (Verify this by examining the cash flows of the hedged borrowing/lending strategy)

The Carry Trade

Higher foreign interest rates imply $F_0 < S_0$ (a **forward discount**).

- As long as exchange rates are unchanged, this is profitable.

Uncovered Interest Rate Parity

Another parity theorem predicts the carry trade *won't* work.

Uncovered interest rate parity (UIP) says:

$$E(S_T) = S_0 \left(\frac{1 + r_{\text{local}}}{1 + r_{\text{foreign}}} \right)^T = F_0$$

Investors will only buy low- (high-) interest-rate currencies if they expect them to appreciate (depreciate).

- **Covered** IRP: the *riskless* (hedged) trade is unprofitable
- **Uncovered** IRP: even the *risky* (unhedged) trade is unprofitable

UIP: Theoretical Implications

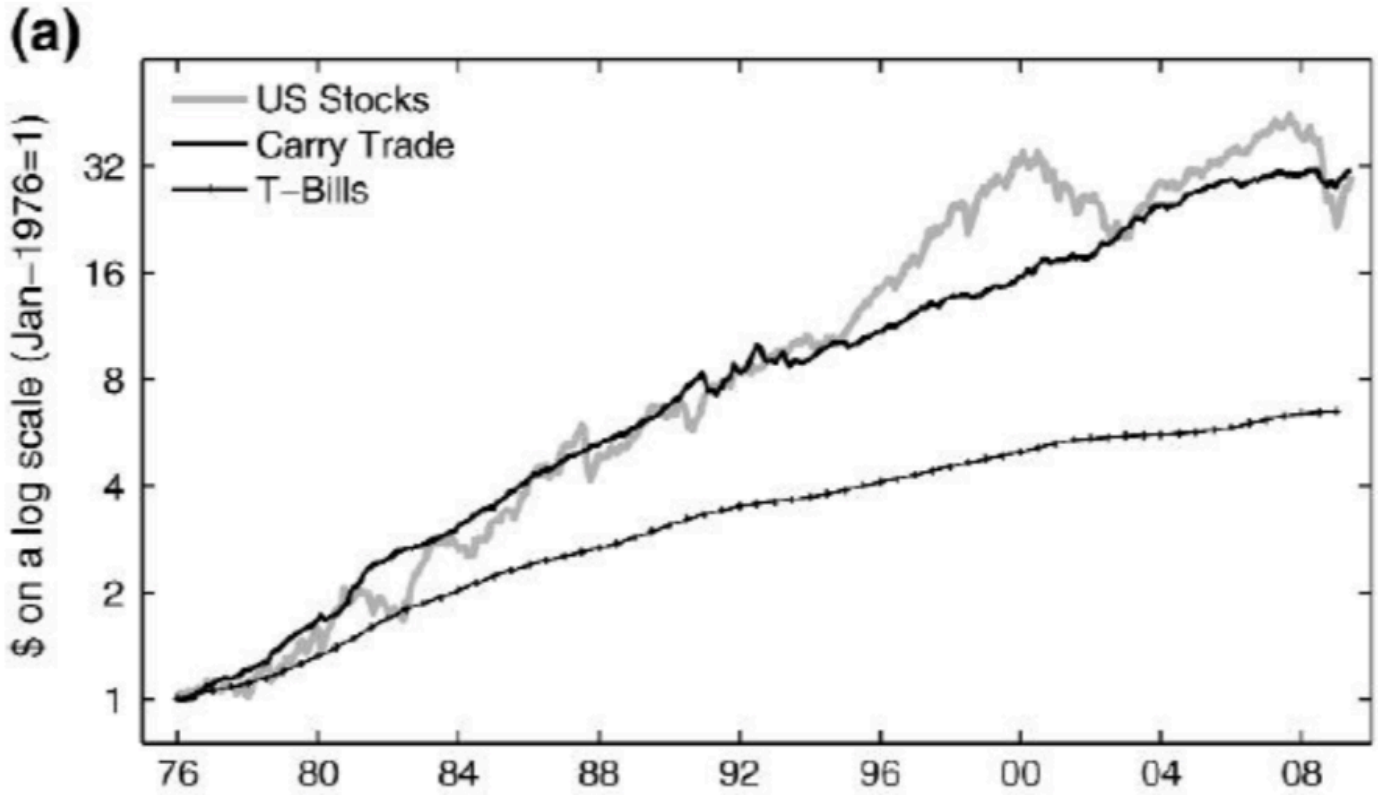
Two equivalent ways to state UIP:

1. **Futures price equals expected spot price**
2. Interest rate differentials are offset by expected changes in spot price:

$$(1 + r_{\text{US}}) = \left(1 + r_{\text{foreign}}\right)(1 + r_{\text{FX}})$$

These imply that the carry trade should only **break even on average**.

Carry Trade Returns



Does UIP Hold? (No.)

UIP predicts that low-interest-rate currencies will *appreciate* relative to high-interest-rate currencies.

Consistently, this result **fails to deliver**:

- Instead, low-interest-rate currencies typically **depreciate** as traders borrow (short) these currencies to lend (buy) in high-yielding currencies

Carry trade profits from **both** the interest rate spread *and* currency appreciation.

Why Does the Carry Trade Work?

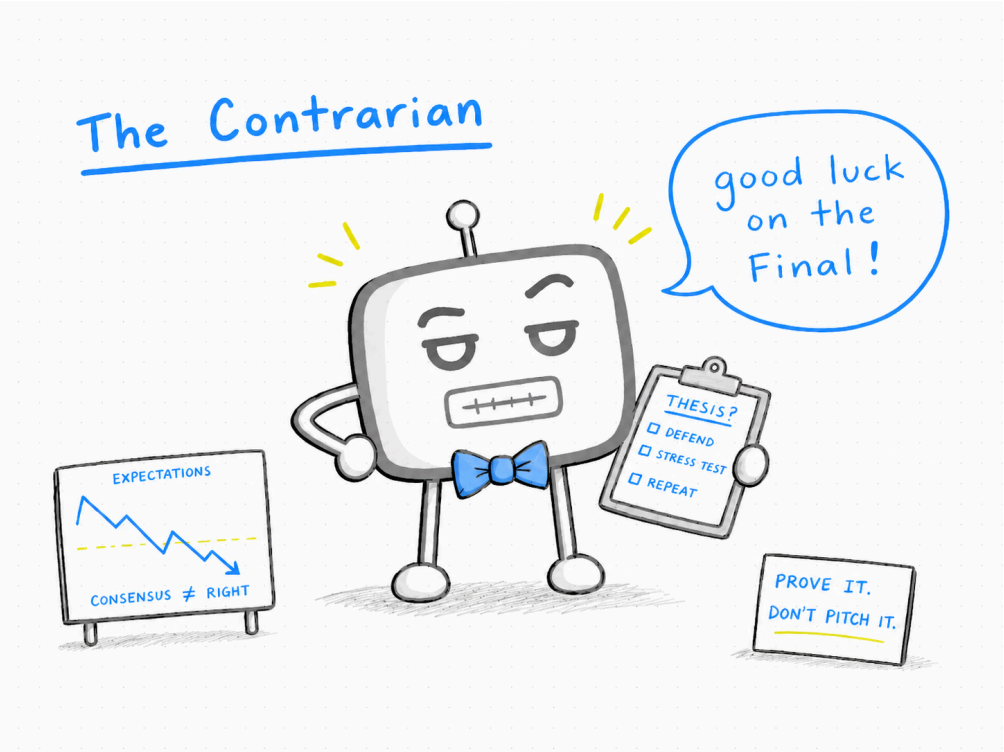
UIP relies on **risk neutrality** — risk aversion may help explain carry returns.

- Burnside (2011): low market loadings, high alpha from a CAPM regression
- Menkhoff, Sarno, Schmeling, and Schrimpf (2011): carry trade does poorly in currency crises

Still largely unexplained — tied to many extreme "tail" events:

- Unwinding during currency crises, financial crises, etc.

Thank You!



Final exam: May 5, 2026; Questions?